

DevelopersHub Corporation

Performance Test Plan

Week 4 – Performance & Load Testing

Application Under Test: automationexercise.com

Tool: Apache JMeter 5.6.3

Date: May 24, 2026

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1. Purpose and Scope

This Performance Test Plan defines the strategy, objectives, test scenarios, and acceptance criteria for load, stress, and spike testing of the e-commerce web application **automationexercise.com**. The goal is to validate that the application can handle concurrent users within acceptable performance thresholds.

2. Objective

- Measure baseline response times for all key pages under normal load.
- Validate system behaviour under 100 concurrent virtual users (Load Test).
- Identify the breaking point under 200 concurrent users (Stress Test).
- Observe system behaviour during a sudden traffic surge of 150 users (Spike Test).
- Collect throughput, error rate, and response time data for each scenario.
- Provide recommendations to improve performance based on results.

3. Scope of Testing

Page / Feature	URL	Method	In Scope
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Home Page	/	GET	Yes
Login Submit	/login	POST	Yes
Products Page	/products	GET	Yes
Signup Page	/signup	GET	Yes
Contact Us Page	/contact_us	GET	Yes

4. Test Environment

Component	Details
Testing Tool	Apache JMeter 5.6.3
Java Version	Java (pre-installed)
OS	Windows 10
Target App	https://automationexercise.com
App Type	E-commerce Web Application
Network	Local WiFi Connection
Test Date	May 24, 2026

5. Test Scenario

ID	Scenario	Virtual Users	Ramp-Up	Loops	Acceptance Criteria
TS-01	Load Test	100	60 sec	3	Avg < 2000ms, Error < 5%
TS-02	Stress Test	200	60 sec	2	Identify breaking point
TS-03	Spike Test	150	5 sec	1	App handles sudden surge

6. JMeter Configuration

HTTP Samplers configured for all scenarios:

Sampler	Protocol	Server	Path	Method
GET Home	https	automationexercise.com	/	GET
POST Login	https	automationexercise.com	/login	POST
GET Products	https	automationexercise.com	/products	GET
GET Signup	https	automationexercise.com	/signup	GET
GET Contact	https	automationexercise.com	/contact_us	GET

Listeners added: View Results Tree, Summary Report, Graph Results

7. KPI

KPI	Target (Load)	Critical (Stress)
Average Response Time	< 2,000 ms	> 5,000 ms = FAIL
Error Rate	< 5%	> 20% = FAIL
Throughput	> 5 req/s	< 2 req/s = WARN
Max Response Time	< 5,000 ms	< 10,000 ms
POST Login Error Rate	< 5%	100% = Critical